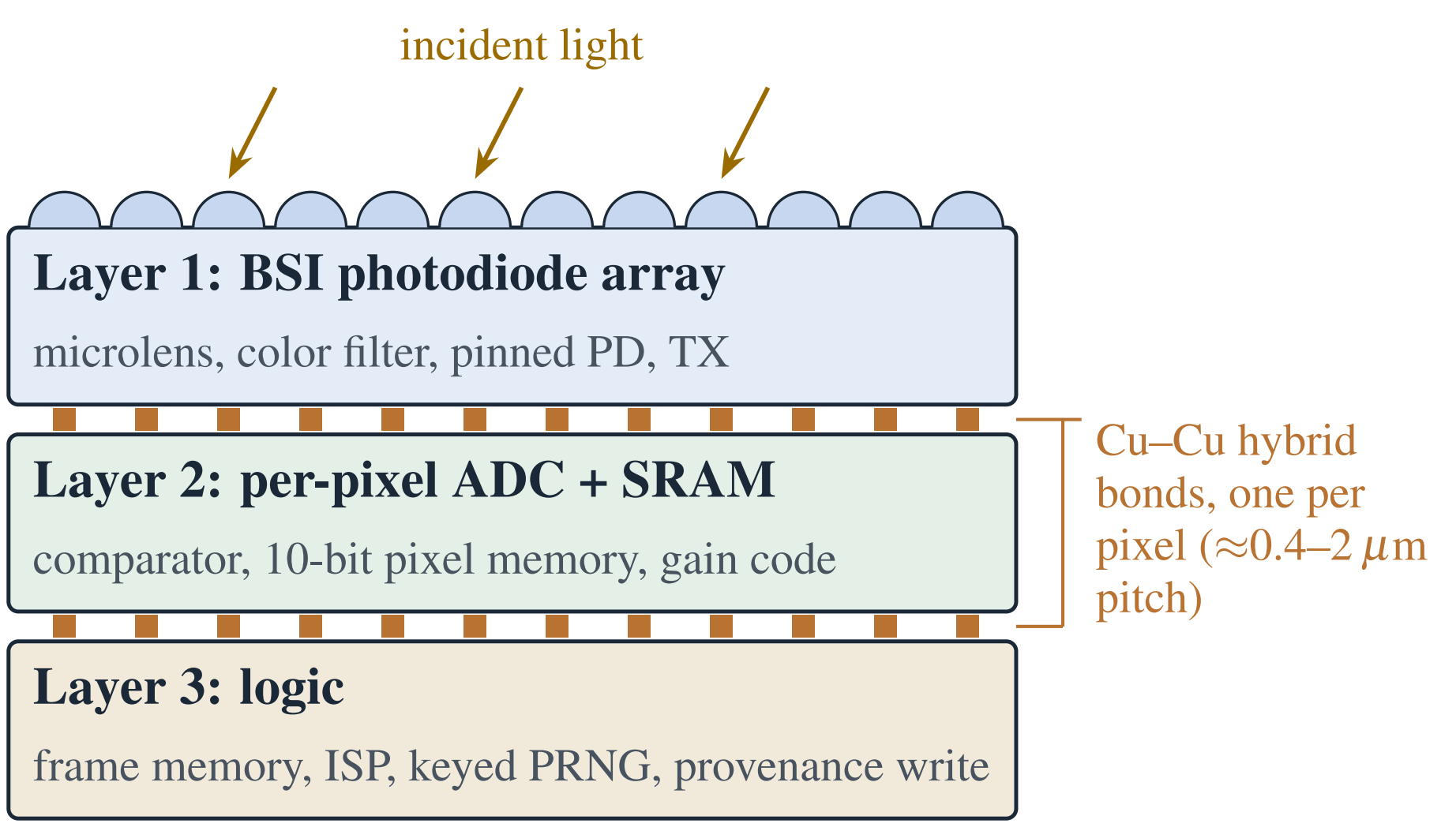
incident light
Layer 1: BSI photodiode array
microlens, color filter, pinned PD, TX
Layer 2: per-pixel ADC + SRAM
comparator, 10-bit pixel memory, gain code
Layer 3: logic
frame memory, ISP, keyed PRNG, provenance write
Cu–Cu hybrid bonds, one per pixel (≈0.4–2 μm pitch)